

NATIONAL CYBERSECURITY LAW

Design Project for VIT — Academic Assignment

Subject

Cybersecurity Law &
Policy

Project Type

Group / Individual Project

Total Marks 100

Marks

Deliverables

Report + Presentation + Flowcharts +
Infographics

Important Notice to Students

This is a comprehensive project assignment. Read all sections carefully before starting. Your team must draft a complete Cybersecurity Law for the fictional country of VIT's. Each part of this project corresponds to a section of the law you will create. Be creative, thorough, and technically accurate.

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PART 1: Introduction

Prepare an overview section that sets the foundation for your cybersecurity law. This section should clearly articulate why cybersecurity legislation is necessary for VIT's.

Purpose of the Law

This law aims to protect citizens, organizations, and government systems from cyber threats and ensure a safe digital environment in VIT's.

Importance of Cybersecurity

As VIT's relies heavily on online banking, e-commerce, AI services, and digital infrastructure, strong cybersecurity laws are essential to protect data and national security.

Challenges Faced by VIT'S

VIT's faces data breaches, ransomware attacks, banking fraud, identity theft, misinformation campaigns, and cyberbullying incidents.

Goals and Expected Outcomes

The law will improve cybersecurity, protect personal data, reduce cybercrime, strengthen public trust, and create a secure digital future for VIT's.

Scenario Context

VIT's is a rapidly growing digital nation with 50 million internet users, online banking, e-commerce platforms, government digital services, smart city infrastructure, AI-based public services, and cloud-based businesses. Recently, VIT's experienced large-scale data breaches, ransomware attacks on hospitals, banking fraud, social media misinformation campaigns, identity theft cases, and cyberbullying incidents.

PART 2: Definitions Section

Define at least 20 important cyber law terms. Your definitions must be precise, legally clear, and relevant to VIT's context. Below are the required terms — you may add more.

Required Terms to Define

#	Term	Definition (To Be Completed by Student)
1	Cybercrime	An illegal activity performed through internet,mobile or computers.
2	Hacker	I person who gains the control of mobile computers or digital devices with or without permission
3	Ethical Hacker	A person who is authorised to check the security of systems
4	Malware	A software which is designed to to damage or gain unauthorised access of system
5	Ransomware	A malware that encrypts data and asks money to decrypt it
6	Phishing	A cyberattack through which sensitive information is gained through websites or fake messages
7	Data Breach	Theft of information
8	Personal Data	Personal information that specifies person's identity for eg:name,email,phone number etc
9	Sensitive Data	Data which contain passwords,financial records,health records
10	Digital Signature	A cryptographic technique by which we can verify authenticity of a person
11	Electronic Evidence	Digital information which can be used as an evidence
12	Identity Theft	Person who uses others identity as its own
13	Artificial Intelligence	Technology for machines to perform activity without human interference
14	Cloud Computing	services such as storage or software on internet
15	Critical Infrastructure	The infrastructure of system and services can be vulnerable to national security,economy, or public safety

16	Cyberbullying	Threatening someone by attaining there personal information
17	Deepfake	Creating any person's AI image,audio or video which seems exactly same
18	Encryption	Process of protecting data by converting normal data into coded form
19	Firewall	A system that monitors incoming and outgoing network traffic
20	Zero-Day Exploit	An attack which takes advantage of software weakness before company fixes it

PART 3: Citizens' Digital Rights

Create a comprehensive Bill of Digital Rights for every citizen of VIT's. For each right, provide a clear explanation of what it means and how it will be enforced.

Required Rights

Right to Privacy

Citizens must be protected from unauthorized collection, use, or sharing of their personal data.

Organizations cannot collect or share personal data without permission. Citizens' information must remain safe and confidential. The NCSA will take action against privacy violations.

Right to Secure Data

All organizations must store and protect citizens' data using industry-standard security measures.

Organizations must protect data using strong security systems. Encryption and firewalls must be used to prevent attacks. Regular security audits will be conducted.

Right to Data Deletion

Citizens may request the complete deletion of their personal data from any organization's systems.

Citizens can request deletion of their personal information. Organizations must remove the data within a fixed time. Failure to comply may result in penalties.

Right to Information

Citizens have the right to know what data is collected about them and how it is used.

Citizens can know what data is collected about them. Organizations must explain how the data is used. Incorrect information can be corrected on request.

Right to Report Cybercrime

Citizens can report cybercrimes without fear of retaliation, and authorities must act within defined timeframes.

Citizens can report cybercrimes without fear or pressure. The government must provide cybercrime helplines and portals. Authorities must act within the defined timeline.

Right to Fair Investigation

Any person accused of a cybercrime is entitled to a fair, transparent, and timely investigation.

Every accused person deserves a fair investigation process. Digital evidence must be collected legally and transparently. Citizens have the right to legal support and appeal.

PART 4: Cyber Offenses and Crimes

Define categories of cybercrime. For each offense, specify a description, real-world example within VIT's context, criminal punishment, and financial fine. Complete the table below.

Crime	Description + Example	Punishment	Fine
Unauthorized Access	Accessing a system without permission. Example: Hacker entered a government server illegally	5 years Imprisonment	₹5 lakh
Data Theft	Stealing confidential digital data. Example: Hospital patient records were stolen.	10 years imprisonment	₹10 lakh
Identity Theft	Using another person's identity online. Example: Fake bank account created using stolen details	5years imprisonment	₹5 lakh
Phishing	Fake emails or links used to steal information. Example: Citizens received fake banking SMS links	4 years imprisonment	₹3 lakh
Ransomware	Malware that locks files and demands money. Example: Hospital systems were encrypted by attackers.	10 years imprisonment	₹20 lakh
Malware Distribution	Spreading harmful software online. Example: Virus spread through fake apps	6 years imprisonment	₹8 lakh
Cyberbullying	Harassing people through digital platforms. Example: Student threatened on social media	3 years imprisonment	₹2 lakh
Online Harassment	Sending repeated abusive online messages. Example: Threatening emails sent to a journalist	3 years imprisonment	₹2 lakh

Financial Fraud	Online cheating for financial gain. Example: Fake UPI payment scam targeted citizens.	7 years imprisonment	₹15 lakh
Misinformation	Spreading false information online. Example: Fake election news created panic.	4 years imprisonment	₹5 lakh
Website Defacement	Changing website content illegally. Example: Government website homepage was hacked.	5 years imprisonment	₹7 lakh
Cyber Terrorism	Cyberattack against national infrastructure. Example: Attack on VIT's power grid system.	Life imprisonment	₹50 lakh

Part 5: Data Protection Framework

Create a comprehensive data protection framework that all organizations operating in VIT's must follow. Your framework must cover the full data lifecycle.

Data Collection Rules

- 1)organisation should collect only required data from students
- 2)personal or sensitive data shouldn't be shared with anyone without students consent
- 3)Data must be collected through secure channels using https or encrypted transmission protocols
- 4)It should be transparent and should be told why the data is collected

Data Storage Rules

- 1)All personal and sensitive data should be encrypted and stored using AES-256 encryption
- 2)regular backup must be done and stored securely

- 3)access to stored data must be restricted
- 4)For administrators their should multi factor authentication

Data Sharing Rules

- 1)data shouldn't be shared until student's consent
- 2)Only necessaryinfo should be shared
- 3)while sharing data should be encrypted using TLS 1.3 or higher
- 4)organisation should maintain a history of data sharing

User Consent Requirements

- 1)students should have ability to withdraw consent anytime
- 2)consent request should be written in understandable language
- 3)consent should be taken for specific tasks as well
- 4)organisation should maintain record's of user's consent

Data Retention Period

- 1)expired data must be archived or deleted
- 2)retention schedule must be documented and reviewed half yearly
- 3)personal data of student should be kept until the purpose is done
- 4)student record must be kept upto 5 years only

Data Deletion Process

- 1)Student should have the right to delete data
- 2)organisation must review deletion request within 15 days
- 3)backup copies of deleted data must removed as well.
- 4)deleted data must be removed by secure data wiping technique
- 5)audit log of deletion must be preserved

Breach Notification Requirements

- 1)Data breach complaint must be reported to relevant authority within 48 hours
- 2)affected student must be reported without delay if there data is at risk
- 3)notification must include nature of breach,affected data and mitigation steps
- 4)organisation must maintain incident response plan for handling data breaches

5)after incident organisation should maintain the steps of action taken to handle the incident for future

PART 6: Corporate Responsibilities

All companies and organizations operating in VIT's must comply with the following cybersecurity responsibilities. Create at least 10 specific, enforceable corporate obligations.

Required Responsibilities (Minimum 10)

Responsibility 1: [cyber security controls]

All companies or organisation should deploy ids,ips,firewall and antivirus software such that they can be protected from harmful networks

Applies to:All organisations

Consequences of non compliance:fine and security audit

Responsibility 2: [regular security audits]

Companies or organisations should conduct cyber security audit in every 6 months

Applies to:IT department

Consequences of non compliance:Fine and mandatory audit

Responsibility 3: [Train employs on cybersecurity]

All employs should be trained about cyber security phishing prevention every three months

Applies to:All employees

Consequences of non compliance:warning and retraining

Responsibility 4: [Protect costumer data]

Organisation must protect costumer's private or sensitive data by encrypting it while storing or sharing

Applies to:Data handlers and administrators

Consequences of non compliance:suspension and fine upto 10 lakhs

Responsibility 5: [incident response plan]

Organisation should regularly test incident response plan for data breaches and cyberattacks

Applies to: security team

Consequences of non compliance: fine upto 10 lakhs

Responsibility 6: [Data breach reporting]

Organisation should report about data breach within 48 hours to the designated authority

Applies to: compliance officer and management

Consequences of non compliance: for delayed reporting fine upto 25 lakhs

Responsibility 7: [access control management]

Access of sensitive or personal data should be given on basis of least privilege

Applies to: it administrator and employees

Consequences of non compliance: guilty should be fined, actions on negligent management

Responsibility 8: [multi factor authentication]

Multi factor authentication should be implemented to all employees and administrators for secure access

Applies to: all employees and administrators

Consequences of non compliance: fine of 1 lakh and implement immediately

Responsibility 9: [software development]

While developing software organisation must use safe coding and run multiple test before launching

Applies to: Developers

Consequences of non compliance: if any damage happen then fine of 5 lakhs and free repair should be done

Responsibility 10: [Disaster recovery]

Organisation should have backup plan to recover from cyber attacks

Applies to: management

Consequences of non compliance: fine of 15 lakhs

Examples to Consider

Maintain security controls | Conduct regular security audits | Train employees on cybersecurity | Report incidents within 72 hours | Protect customer information | Appoint a Data Protection Officer | Conduct annual penetration testing | Implement access control policies | Encrypt sensitive data | Maintain incident response plans

PART 7: Government Responsibilities

Specify how government agencies of VIT's must protect national interests in cyberspace. Include concrete mechanisms for each area.

Protect National Infrastructure

The government will protect banks, hospitals, power plants, and transport systems using strong cybersecurity systems and regular security monitoring.

Monitor Cyber Threats

Government agencies will use threat intelligence systems and AI tools to detect cyberattacks and suspicious activities in real time.

Investigate cybercrimes

Special cybercrime units and digital forensic teams will investigate cybercrimes according to legal procedures and collect digital evidence.

International Coordination

VIT's will cooperate with other countries and organizations like INTERPOL to share cyber threat information and investigate international cybercrimes.

Awareness Programs

The government will conduct cybersecurity awareness campaigns in schools, colleges, companies, and social media platforms to educate citizens.

PART 8: Cyber Incident Reporting Framework

Design a step-by-step process for how cyber incidents are reported, investigated, and resolved in VIT's. You must also create a flowchart (as a separate deliverable).

Reporting Process Steps

Step	Stage	Description (Student to Complete)
1	Incident Detected	The organization detects a cyberattack and informs the internal security team immediately.
2	Internal Investigation	The cybersecurity team checks logs, identifies affected systems, and collects evidence.
3	Report to Authority	The incident must be reported to the NCSA within 24 hours.
4	Risk Assessment	Authorities analyze the impact and determine the severity of the attack.
5	Mitigation	Security teams isolate infected systems and stop the attack from spreading.
6	Recovery	Systems and data are restored using backups and security patches.
7	Final Report	A final investigation report is prepared with actions taken and lessons learned.

Deliverable Reminder

You must submit a separate FLOWCHART showing this complete process. Use tools like Canva, draw.io, Lucidchart, or PowerPoint SmartArt to create a professional diagram.



PART 9: National Cyber Security Authority (NCSA)

Design the organizational structure, responsibilities, and powers of the NCSA — the primary government agency responsible for cybersecurity in VIT's.

Organizational Structure

The NCSA will be headed by a Director General with departments for threat intelligence, digital forensics, incident response, and public awareness.

Core Responsibilities

The NCSA will monitor cyber threats, investigate cybercrimes, create cybersecurity policies, and conduct awareness programs.

Legal Powers

The NCSA can investigate cyber incidents, collect digital evidence, order security audits, and shut down harmful websites.

Investigation Process

The NCSA receives complaints, collects evidence, performs digital forensic analysis, identifies suspects, and submits reports to the court.

Departments

Main departments include Threat Intelligence Unit, Digital Forensics Lab, Incident Response Team, Cybercrime Investigation Unit, and Public Awareness Division.

PART 10: AI and Emerging Technology Regulations

Create specific cybersecurity regulations for emerging technologies used in VIT's. For each technology, identify key risks and write enforceable rules.

Artificial Intelligence

Key Risks

- 1) bias decision making can happen in automated system
- 2) Ai generated email or social engineering attack
- 3) Ai generated misleading messages
- 4) Accessing student or faculty data using ai

Regulations

- 1) While judging or checking papers ai shouldn't be used
- 2) Any information generated through Ai should be checked by human before distributing
- 3) All Ai systems where sensitive data should implement encryption
- 4) ai generated content should be disclosed by users

Deepfakes

Key Risks

- 1) creation of fake audios or videos of student or faculty
- 2) Impersonation
- 3) risk of Damage of reputation
- 4) social engineering attacks using fake voices

Regulations

- 1) Ai video should be labelled
- 2) Deepfake video intended to harm anyone. immediate disciplinary action should be taken
- 3) Ai audio should be labelled
- 4) creation of deepfakes should be prohibited

IoT Devices

Key Risks

- 1)data can be intercepted from sensors and smart devices
- 2)insecure iot devices can spread malware
- 3)weak passwords can lead to compromise of device

Regulations

- 1)strong passwords recommended
- 2)Every device should approved by it department
- 3)iot device network should be speerated from university system

Cloud Computing

Key Risks

- 1)Misconfiguration of cloud services can lead to data breach
- 2)unauthorised access to cloud stored information
- 3)loss of data availability during loss of service
- 4)sensitive student data can get leaked

Regulations

- 1)multi factor authentication should be mandatory to cloud services
- 2)access permissions must follow the principle of least privilege
- 3)backup of the data should be done regularly
- 4)encryption should be done even while transferring or at rest

Digital Identity Systems

Key Risks

- 1)Identity theft or credential can be compromised
- 2)unauthorised access to college systems
- 3)phishing attacks through user credentials
- 4)privacy at risk including personal information

Regulations

- 1)strong passwords recommended

- 2)identity verification must be mandatory before granting access**
- 3)every account must be reviewed every day and which account has no action must be disabled**
- 4)personal identity data must kept private according to privacy policy**

Autonomous Systems

Key Risks

- 1)unauthorised control of autonomous devices and systems**
- 2)software vulnerabilities causing unsafe operations**
- 3)decision making can be affected due to data manipulation**
- 4)system failure can affect college operations**

Regulations

- 1)before deployment autonomous system should be tested**
- 2)software patches and security updates should be done regularly**
- 3)every autonomous system which is deployed must be monitored regularly**
- 4)emergency shutdown must be available**

PART 11: Critical Infrastructure Protection

Create sector-specific cybersecurity requirements for each of the following critical infrastructure types in VIT's. Requirements should be practical and technically grounded.

Sector	Cybersecurity Requirements (Student to Complete)
Banks & Financial	1)multi factor authentication should be implemented for costumers and employees 2)employees should always monitor system to detect fraud. 3)encryption of all transactions and and costumer data 4)there should be always a disaster recovery plan
Hospitals & Healthcare	1)Educate employees on phishing and social engineering attacks 2)access of patient records to be given to authorised person using role based access control 3)medical systems should be regularly patched 4)conduct ransomware protection and data backups
Airports & Aviation	1)secure air traffic and communication system 2)strong authentication for airport employees 3)keep softwares of aviation system updated with security patches 4)separate the public wifi and office networks
Railways & Transport	1)conduct cybersecurity audit for oerating sytems 2) IDS should be implemented to monitor network 3)encrypt communication between stations and trains
Power Plants & Energy	1)strict security over access to engineers and operators 2)keep softwares updated with security patches 3)monitor system 24/7 over cyber attacks
Government Data Centers	1)encrypt sensitive data over transit and when stored 2)monitor system 24/7 over cyber attacks 3)maintain offline backups 4)multifactor authentication for all government employees 5)use firewalls and IDS to detect and protect system from harmful networks

PART 12: Awareness and Education

Design a National Cybersecurity Awareness Campaign for VIT's. Your campaign must reach all segments of society through age-appropriate and effective programs.

School Programs (Age 6-18)

Schools will teach safe internet usage, password security, cyberbullying prevention, and online privacy through interactive activities and workshops.

College & University Programs

Colleges will offer cybersecurity courses, ethical hacking labs, certifications, and cyber awareness seminars for students.

Corporate Training Programs

Companies must provide cybersecurity training to employees every six months and conduct phishing awareness tests.

Social Media Campaigns

The government will share cybersecurity tips, videos, posters, and infographics on social media platforms to reach citizens.

Public Awareness Workshops

Free workshops will be organized in communities, libraries, and public centers to educate people about cyber threats and online safety.

PART 13: International Cooperation

Define how VIT's will collaborate with other countries and international organizations to combat cybercrime that crosses national borders.

Cybercrime Investigations

VIT's will cooperate with foreign cybercrime agencies to investigate international cyberattacks and share evidence legally.

Information Sharing Agreements

VIT's will exchange cyber threat information and security alerts with trusted countries and international organizations.

Joint Operations

VIT's may conduct joint cybercrime investigations and operations with foreign law enforcement agencies when required.

Extradition Requests

Cybercriminals hiding in other countries may be extradited to VIT's under international agreements and legal procedures.

Threat Intelligence Exchange

VIT's will regularly share threat intelligence, malware reports, and attack indicators with global cybersecurity partners.

PART 14: Case Studies

Create 3 original fictional cybercrime cases set in VIT's. Each case must be realistic, detailed, and demonstrate how your law would be applied.

Case Study 1: [Government Database Breach]

Crime Category	Unauthorized Access / Data Breach
What Happened	A hacker gained unauthorized access to the national citizen database and stole personal information of 50,000 citizens
Law Violated	Unauthorized Access, Data Theft, Right to Privacy Violations
Investigation Process	The NCSA traced the hacker using server logs, collected digital evidence, and arrested the suspect.
Punishment Given	7 years imprisonment and ₹10 lakh fine.
Lessons Learned	Government agencies implemented stronger encryption and multi-factor authentication.

Case Study 2: [Fake Banking SMS Scam]

Crime Category	Financial Fraud / Phishing
What Happened	Cybercriminals sent fake banking messages and tricked citizens into sharing account details.
Law Violated	Phishing, Identity Theft, Financial Fraud
Investigation Process	The NCSA tracked the fake website, analyzed transaction records, and arrested the criminals.
Punishment Given	5 years imprisonment and ₹15 lakh fine.
Lessons Learned	Banks improved customer awareness and introduced stronger account security measures.

Case Study 3: [Hospital Ransomware Attack]

Crime Category	Ransomware / Cyber Terroris
What Happened	Attackers infected a major hospital network with ransomware, making patient records inaccessible.
Law Violated	Ransomware Attack, Critical Infrastructure Attack
Investigation Process	The NCSA isolated infected systems, performed digital forensics, and identified the attackers.
Punishment Given	10 years imprisonment and ₹20 lakh fine
Lessons Learned	Hospitals adopted regular backups, cybersecurity training, and continuous security monitoring.

PART 15: Future Recommendations

Based on your research, suggest forward-looking improvements and additions to VIT's cybersecurity law for the next 10 years. Consider how technology will evolve and what new threats will emerge.

AI Security (2025-2035)

- 1) Establish regulations for responsible use of Ai
- 2)every ai generated video should be labelled to decrease misuse of ai
- 3)ai systems should be regularly monitored and it should be transparent
- 4)for decision making Ai should not be used human should interfere In critical decision making

Quantum Computing

- 1)begin quantam resistant cryptography to protect sensitive information
- 2)reagularly check national infrastructure for future weaknessess to future cyber attack
- 3)support research and development in post quantam security technologies
- 4)banks healthcare institutions and government agencies shouldupgrade encryption standards before big quantam attack threat appears

Privacy Protection

- 1)Heavy penalties should be implemented on misuse and breach of personal data
- 2)to protect biometric and smart device data use high security standards
- 3)Give every citizen control over there personal data such that only they can access correct and delete data
- 4)organisation should take only necessary data and should take the responsibility of it not share

Digital Identity

- 1)Implement secure identity for system such that it protects the privacy
- 2)implement multi factor authentication and strong encryption to prevent identity theft
- 3)store minimal sensitive data to protect from breaches
- 4)digital identity should be accessible and transparent

Cyber Warfare

- 1)clearly define state sponsored cyber attacks
- 2)invest more on cyber security technologies and threat intelligence

- 3) strengthen great cooperation when investigating international cyber attacks
- 4) create a framework for responding to a cyber attack against critical infrastructure

Evaluation Criteria

Your project will be graded based on the following criteria. Review this carefully to maximize your marks.

Done	Deliverable	Requirements
☐	Project Report	Minimum 25 pages, wellformatted, all 15 parts completed, diagrams included
☐	PowerPoint Presentation	15 to 20 slides, covering all major sections of the law
☐	Flowchart: Cybercrime Reporting	Shows complete incident reporting process from detection to resolution

[]	Flowchart: Investigation Process	Shows NCSA investigation steps from complaint to judgment
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Criteria	Max Marks	Marks Obtained
Research Quality — depth and accuracy of information, use of real-world examples	20	
Law Structure — completeness, logical organization, and legal clarity of the law	20	
Innovation — originality of solutions, creative approach to cybersecurity challenges	15	
Technical Accuracy — correctness of technical details, cybersecurity concepts, and procedures	15	
Presentation — quality of the PowerPoint presentation and oral delivery	10	
Diagrams & Flowcharts — clarity, accuracy, and professionalism of visual deliverables	10	
Case Studies — realism, detail, and correct application of the law	10	
TOTAL	100	

Deliverables Checklist

Before submitting, ensure you have completed all required deliverables:

Done	Deliverable	Requirements
[]	Flowchart: Data Breach Response	Shows data breach response procedure for organizations

[]	Infographic: Citizens' Rights	Visual summary of all citizens' digital rights
[]	Infographic: Cybercrime Categories	Visual taxonomy of all defined cybercrimes with penalties
[]	Infographic: Security Framework	Visual overview of the data protection and corporate security framework

Good Luck!

This project challenges you to think like a lawmaker, a technologist, and a citizen. Research realworld cybersecurity laws (India's IT Act, GDPR, Budapest Convention) for inspiration. Be creative, be thorough, and make VIT's digital future secure!